

Task 1: ML Classification – Customer Churn Prediction

Internspark AI Internship

1. GitHub Repository

<https://github.com/zom1xo/churn-classification>

2. Notebook

Direct link to .ipynb file in the repo

3. Summary

Built a binary classification model to predict customer churn using the Telco Customer Churn dataset. Compared Logistic Regression and Random Forest with stratified 5-fold cross-validation. Evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. Random Forest achieved the higher ROC-AUC. Top predictive features: tenure, contract type, and monthly charges. Model saved for Task 4 (Responsible AI & SHAP/LIME).

4. Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
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Logistic Regression	0.7991	0.6426	0.5481	0.5916	0.8402
Random Forest	0.7956	0.6463	0.5080	0.5689	0.8358

Logistic Regression slightly outperformed Random Forest with a higher ROC-AUC (0.8402 vs 0.8358). Both models struggled with recall for the churn class due to class imbalance (~27% churn). Top predictive features: tenure, contract type, monthly charges, and total charges. Model saved for Task 4 (Responsible AI & SHAP/LIME).

5. Key Visuals

- ROC curves (both models on one plot)
- Confusion matrices side-by-side
- Feature importance bar chart

6. Environment

- Python 3.x, Google Colab
- Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, joblib